

Signed numbers: addition and subtraction

Initial assessment • Work out all four questions. Show your working.

1. $-2 + 9$

2. $6 - 11$

3. $-7 - (-10)$

4. $5 + (-9) - (-6)$

Signed numbers: addition and subtraction

Rules recap • Start at the first number.

Add a positive number

Move right: the value increases.

$$-4 + 7 = 3$$

Add a negative number

Move left: the value decreases.

$$4 + (-7) = 4 - 7 = -3$$

Subtract a positive number

Move left: the value decreases.

$$3 - 8 = -5$$

Subtract a negative number

Move right: subtracting a negative is adding a positive.

$$6 - (-3) = 6 + 3 = 9$$

In $6 - (-3)$, the first $-$ means subtract;
the $-$ inside brackets belongs to the number -3 .

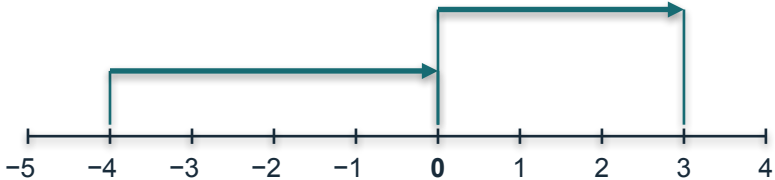
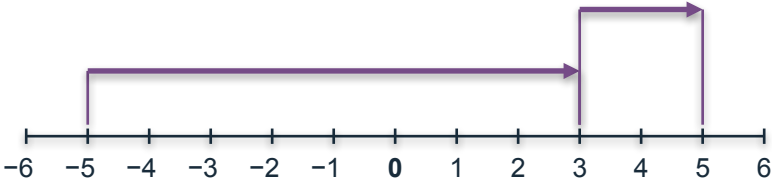
To subtract a number, add its opposite. Keep the first number unchanged.

Two negative numbers do not automatically give a positive answer:
 $-4 + (-3) = -7$.

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Worked demo • Follow each calculation.

Adding a negative moves left: $4 + (-7) = 4 - 7 = -3$.
Subtracting a negative moves right: $6 - (-3) = 6 + 3 = 9$.

	Thread 1: Guided	Thread 2: Core	Thread 3: Depth
The Full Question	$-4 + 7$	$6 - (-3)$	$-5 + 8 - (-2)$
Read the Signs	Start at -4 ; add positive 7 .	Start at 6 ; subtract negative 3 .	Add positive 8 , then subtract negative 2 .
Rewrite the Calculation	$-4 + 7$ is already an addition.	$6 - (-3) = 6 + 3$.	$-5 + 8 + 2$.
Calculate	Move 4 right to 0, then 3 right: 3. $+4$ then $+3$: total $+7$ 	$6 + 3 = 9$. subtract $-3 =$ add 3 	$-5 + 8 = 3$; $3 + 2 = 5$. $+8$, then $+2$ 
Check the Answer	$3 - 7 = -4$. Answer: 3 .	$9 + (-3) = 6$. Answer: 9 .	$5 + (-2) = 3$; $3 - 8 = -5$. Answer: 5 .

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Step-by-step practice 1 • Show your working.

	Thread 1: Guided	Thread 2: Core	Thread 3: Depth
The Full Question	$-6 + 9$	$4 + (-7)$	$-8 + 5 - (-4)$
Read the Signs	Identify the operation signs and the signs belonging to the numbers.	Identify the operation signs and the signs belonging to the numbers.	Identify the operation signs and the signs belonging to the numbers.
Rewrite the Calculation	Replace subtraction by addition of the opposite where useful.	Replace subtraction by addition of the opposite where useful.	Replace subtraction by addition of the opposite where useful.
Calculate	Work from left to right. Show each result.	Work from left to right. Show each result.	Work from left to right. Show each result.
Check the Answer	Undo the last operation to check. Write the final answer.	Undo the last operation to check. Write the final answer.	Undo the last operation to check. Write the final answer.

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Step-by-step practice 2 • Show your working.

	Thread 1: Guided	Thread 2: Core	Thread 3: Depth
The Full Question	$3 - 8$	$-5 - (-9)$	$7 - (-2) + (-12)$
Read the Signs	Identify the operation signs and the signs belonging to the numbers.	Identify the operation signs and the signs belonging to the numbers.	Identify the operation signs and the signs belonging to the numbers.
Rewrite the Calculation	Replace subtraction by addition of the opposite where useful.	Replace subtraction by addition of the opposite where useful.	Replace subtraction by addition of the opposite where useful.
Calculate	Work from left to right. Show each result.	Work from left to right. Show each result.	Work from left to right. Show each result.
Check the Answer	Undo the last operation to check. Write the final answer.	Undo the last operation to check. Write the final answer.	Undo the last operation to check. Write the final answer.

Signed numbers: addition and subtraction

Independent practice 1 • Choose a column. Show your working and check your answers.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. $-3 + 8$	1. $5 - (-2)$	1. $-4 + 7 - (-3)$
2. $2 - 7$	2. $7 + (-9)$	2. $6 - 9 + 2$
3. $-6 + 4$	3. $-4 + (-6)$	3. $-5 + (-3) + 10$
4. $-2 - 5$	4. $-8 - (-3)$	4. $8 - (-2) - 13$
5. $-9 + 9$	5. $-5 - (-5)$	5. $-7 - (-4) + 1$
6. $4 - 10$	6. $0 + (-7)$	6. $3 + (-8) - (-5)$

Signed numbers: addition and subtraction

Independent practice 2 • Choose a column. Show your working and check your answers.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. $-4 - 6$	1. $9 - (-4)$	1. $-6 + 9 + (-4)$
2. $-11 + 11$	2. $-6 + (-7)$	2. $4 - (-6) + (-10)$
3. $6 - 15$	3. $0 - (-9)$	3. $7 - 10 - (-5)$
4. $-7 + 10$	4. $3 + (-8)$	4. $-8 - (-2) - 1$
5. $-8 + 3$	5. $-3 - (-8)$	5. $-9 + 5 - (-7)$
6. $5 - 12$	6. $-10 - (-6)$	6. $5 + (-12) + 3$

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Independent practice 3 • Choose a column. Show your working and check your answers.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. $-8 + 12$	1. $-9 + (-2)$	1. $10 - 14 + (-2)$
2. $-5 - 4$	2. $8 - (-5)$	2. $6 + (-9) - (-3)$
3. $-10 + 6$	3. $-7 - (-4)$	3. $-2 - (-8) + 5$
4. $9 - 14$	4. $6 + (-11)$	4. $-3 + (-6) - (-4)$
5. $7 - 16$	5. $-4 - (-10)$	5. $9 - (-3) + (-7)$
6. $-12 + 12$	6. $-12 - (-7)$	6. $-11 + 8 - 2$

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Independent practice 4 • Choose a column. Show your working and check your answers.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. $-12 + 7$	1. $0 - (-6)$	1. $-12 + 7 + 8$
2. $11 - 18$	2. $-11 - (-8)$	2. $4 - (-5) + (-12)$
3. $-6 - 8$	3. $5 + (-13)$	3. $-6 - (-9) + (-3)$
4. $-9 + 14$	4. $-7 - (-12)$	4. $5 - 13 - (-2)$
5. $-15 + 15$	5. $12 - (-3)$	5. $8 + (-11) - 4$
6. $8 - 13$	6. $-4 + (-9)$	6. $-10 + 6 - (-7)$